Mastering Fabric Data Agents:
From Setup to Success



Thank you, partners















































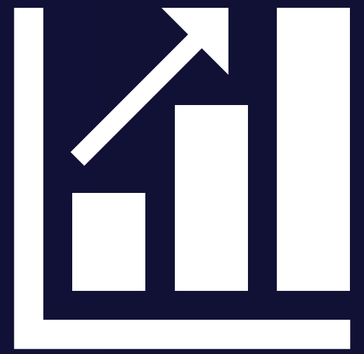
Let’s Define Success



Reliable answers



Secured



Well adopted

Personas	Customer	Financial Advisors	Portfolio Managers	...
Modern Technology	Co-pilot / Chatbot		Predictions	
Data Management Platform	 Fabric			
Platforms	Digital Experience	CRM & ERP	Document Management	...

Marc Lelijveld

Architect & Technical Advocate
Macaw Netherlands

Microsoft® MVP Most Valuable Professional

linkedin.com/in/MarcLelijveld

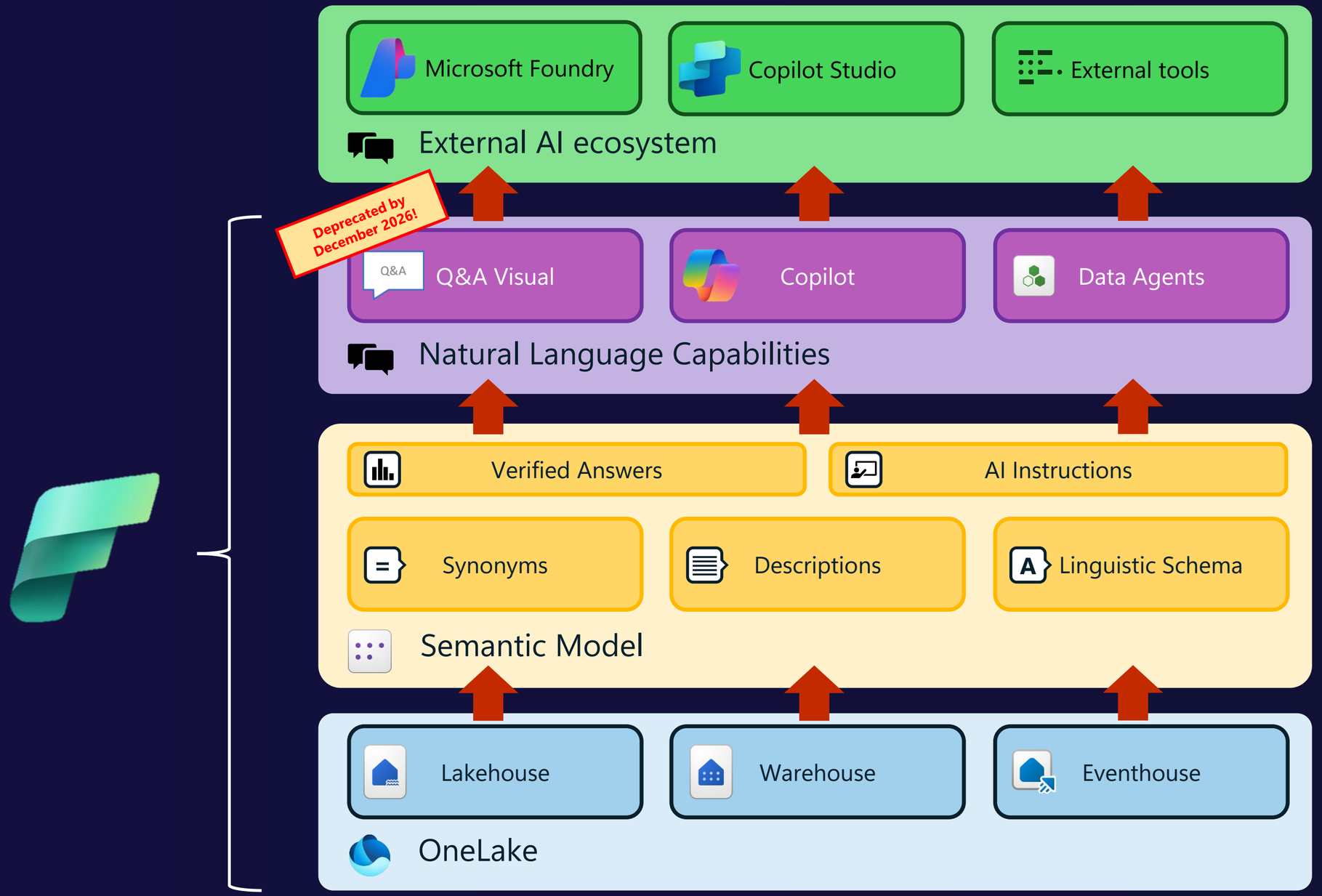
Data-Marc.com



Exploring solution areas
What are the alternatives to copilot?

What can you conclude from the orders placed versus the orders shipped?

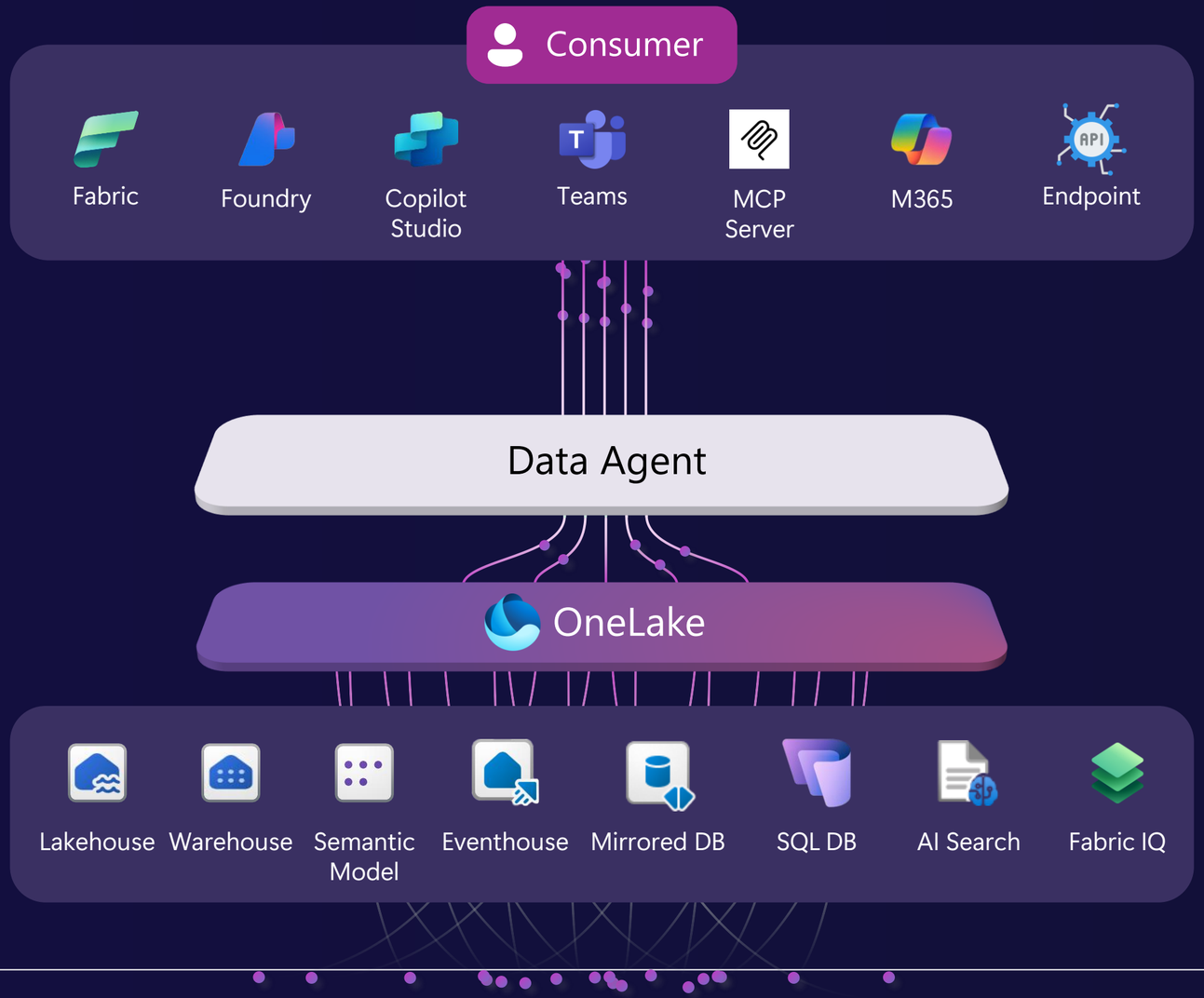



Microsoft Foundry
Copilot Studio
External tools
External AI ecosystem
Deprecated by December 2026!
Q&A
Q&A Visual
Copilot
Data Agents
Natural Language Capabilities
Verified Answers
AI Instructions
Synonyms
Descriptions
Linguistic Schema
Semantic Model
Lakehouse
Warehouse
Eventhouse
OneLake

Prerequisites for Data Agents

- Fabric paid capacity (any size)
- Fabric Data Agent tenant setting is enabled
- Copilot tenant switch is enabled
- A valid source (Warehouse, Lakehouse, Power BI semantic model or KQL Database)
- Power BI semantic models via XMLA endpoints tenant switch is enabled.

Fabric Data Agents



A **virtual analyst** that allows users to interact with and gain insights from enterprise data in OneLake

New Consume your data agents in **M365 Copilot** and as **remote MCP Server in VS Code**

New Data agents now support unstructured data through **Azure AI Search**

New Data agents now support **Fabric IQ Workload through Ontology**

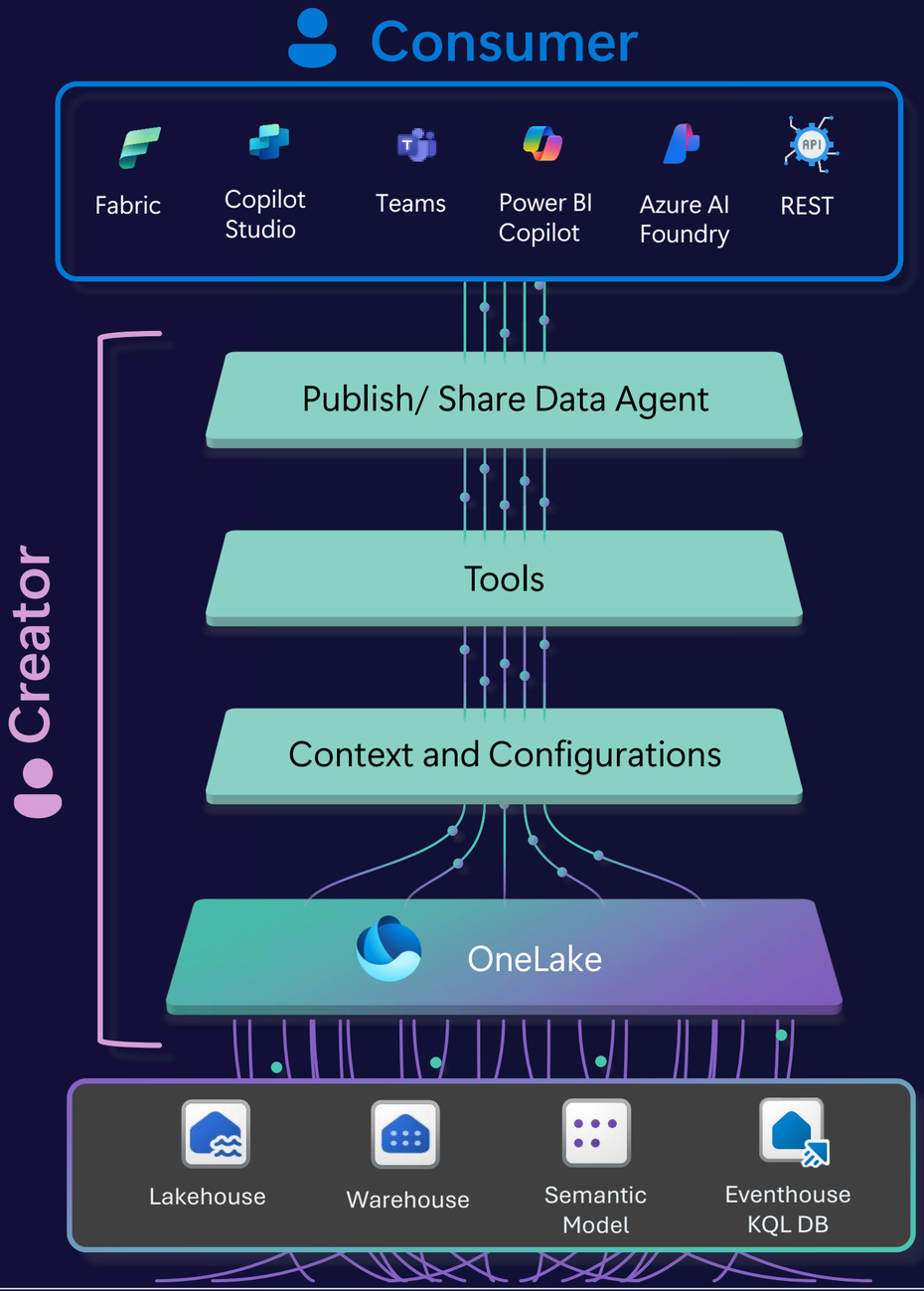
Introducing Fabric Data Agents

The **conversational Data Agent** allows users to interact with data naturally, enhancing accessibility and usability.

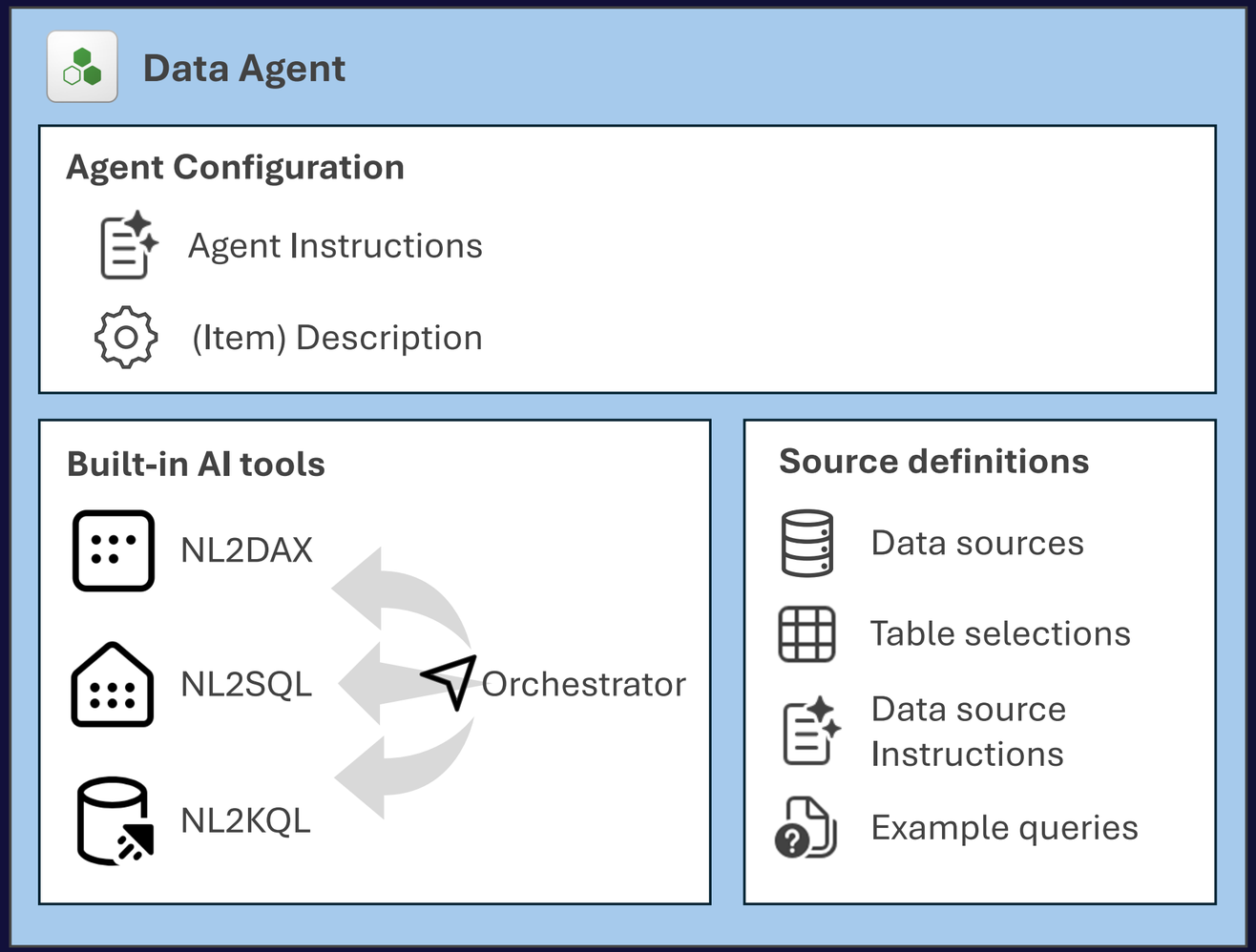
Improved chat canvas for creators with **new debugging capabilities**, making it easier to understand and refine responses.

Seamlessly reason over multiple data sources, including **Lakehouses, warehouses, semantic models and Eventhouse KQL databases**.

Your Data Agent can be consumed inside and outside of Fabric. Stay tuned for **upcoming integrations** with Copilot Studio, Teams, Azure AI Foundry and your own custom applications.



Data Agent building blocks



Creator configurations

Instructions (schema)

Verified answers

Agent Instructions

Data Source Instructions

Example Queries

Intended Use Cases:

- Specify which data sources to route for specific questions
- Guide how the agent interprets user intent

What to include:

- Map terminology to user intent
- Define steps for handling key questions
- Set response style and tone

Intended Use Cases:

- Provide clarity on how to use each source
- Guide table selection for topics
- Ensure correct query logic

What to include:

- Descriptions of key tables & columns
- Recommended table usage for certain topics
- Example column values and filter patterns

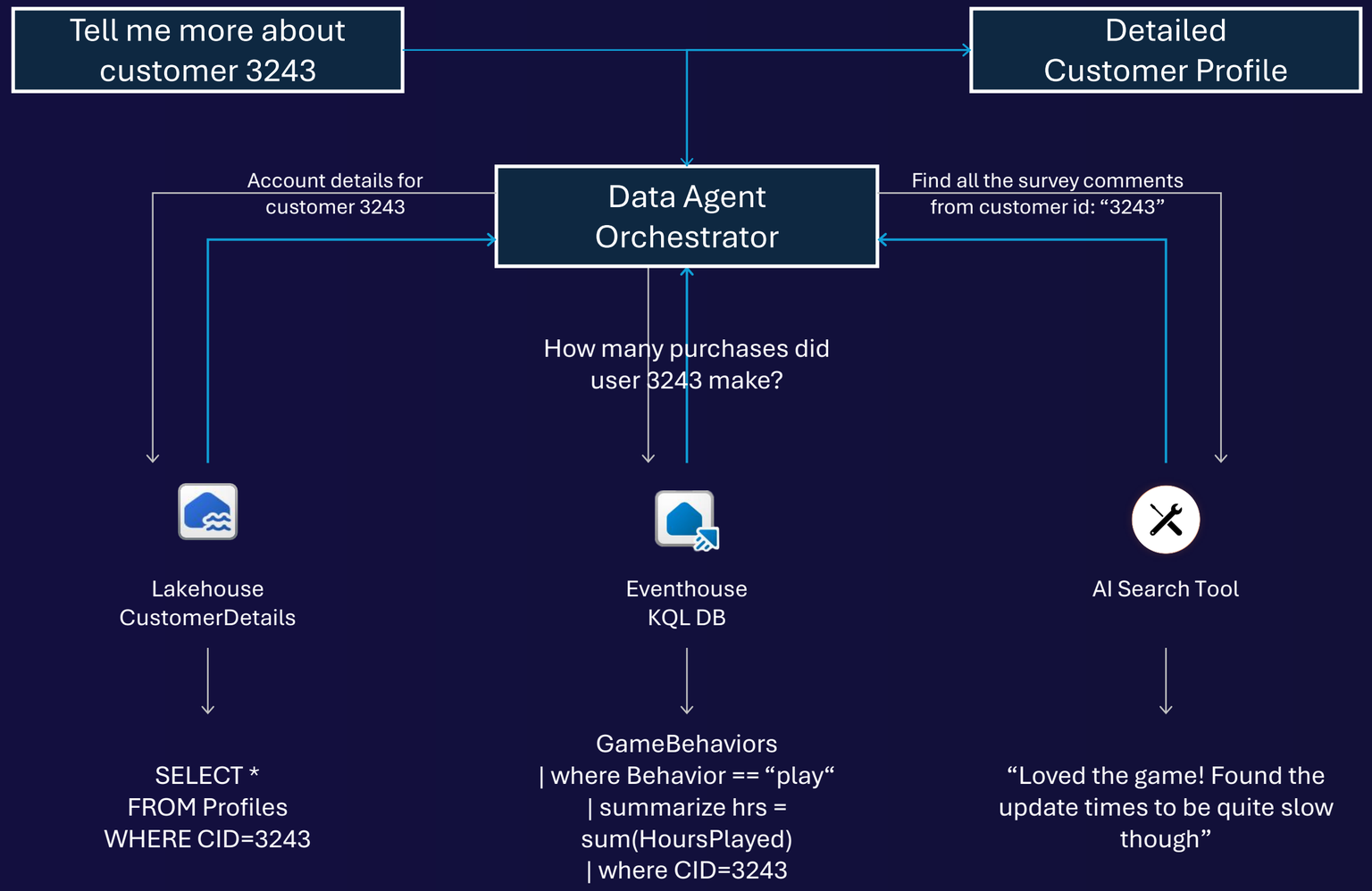
Intended Use Cases:

- Reflect advanced logic
- Examples should highlight join patterns or expected values

What to include:

- Representative set of questions a consumer may ask

The Data Agent flow



1. The data agent delegates questions to specific tools or data sources.

2. Data Sources are specialized at taking a natural language question and generating a response – query or result.

3. Data Sources & Tools supply their response back to the orchestrator – allowing it to process the information and think about the next step or return the summarized result.

Different types of supported sources



Lakehouse



Warehouse



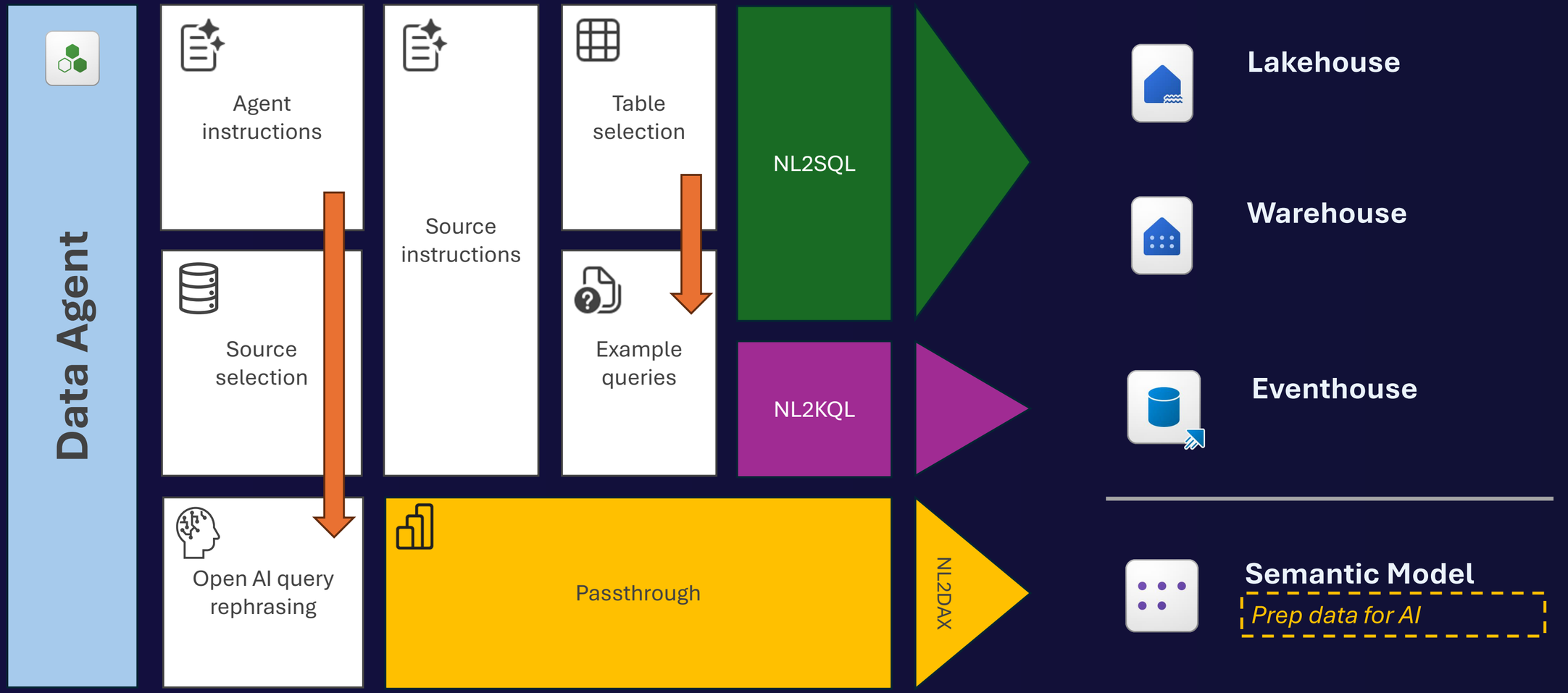
Eventhouse



Semantic Model

And related sources (like mirrored sources)

Data Agent Orchestrator - Inner workings

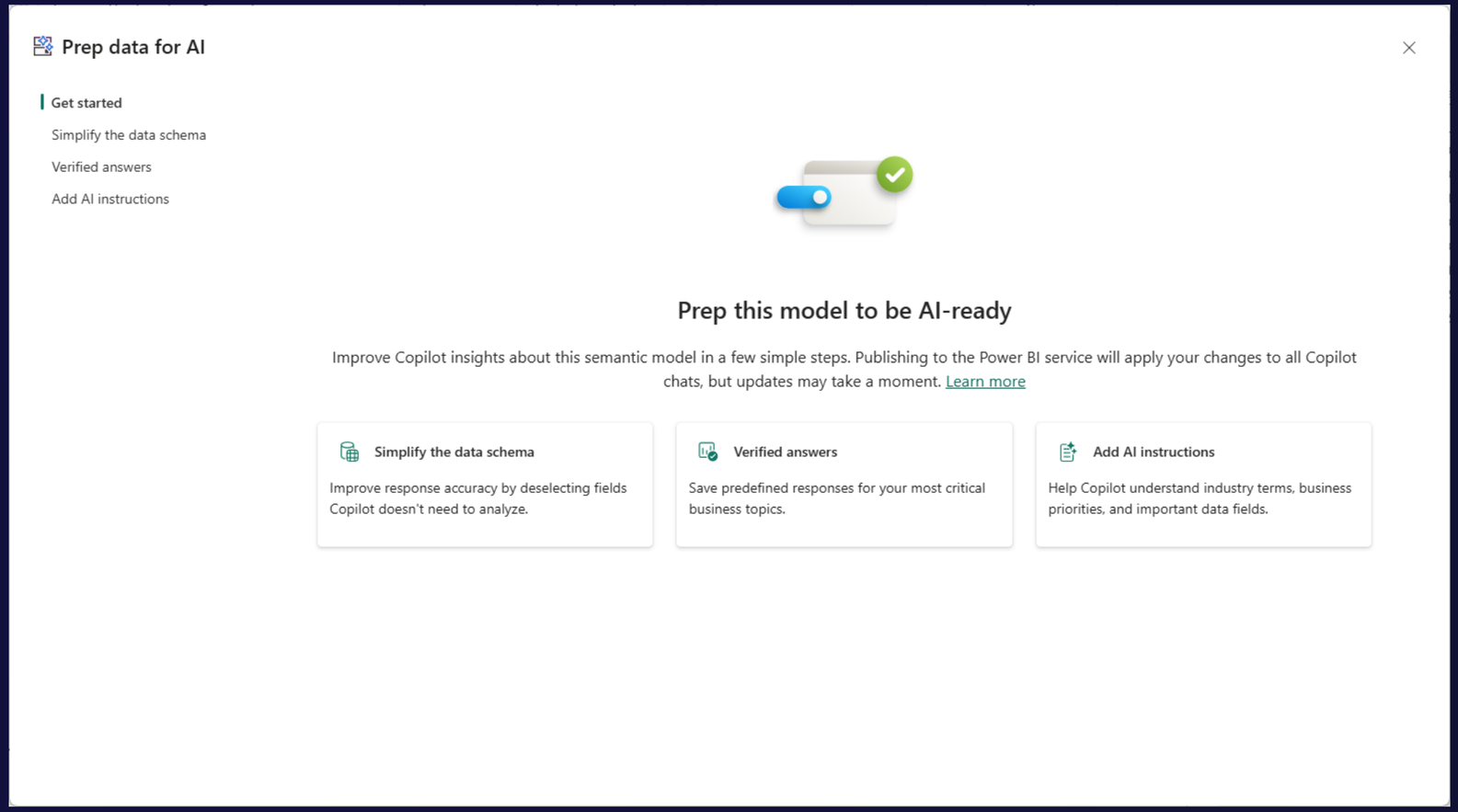


Semantic model only

Semantic models: Prep data for AI

Three main pillars

- Simplify schema
- Verified answers (links to visuals)
- AI Instructions

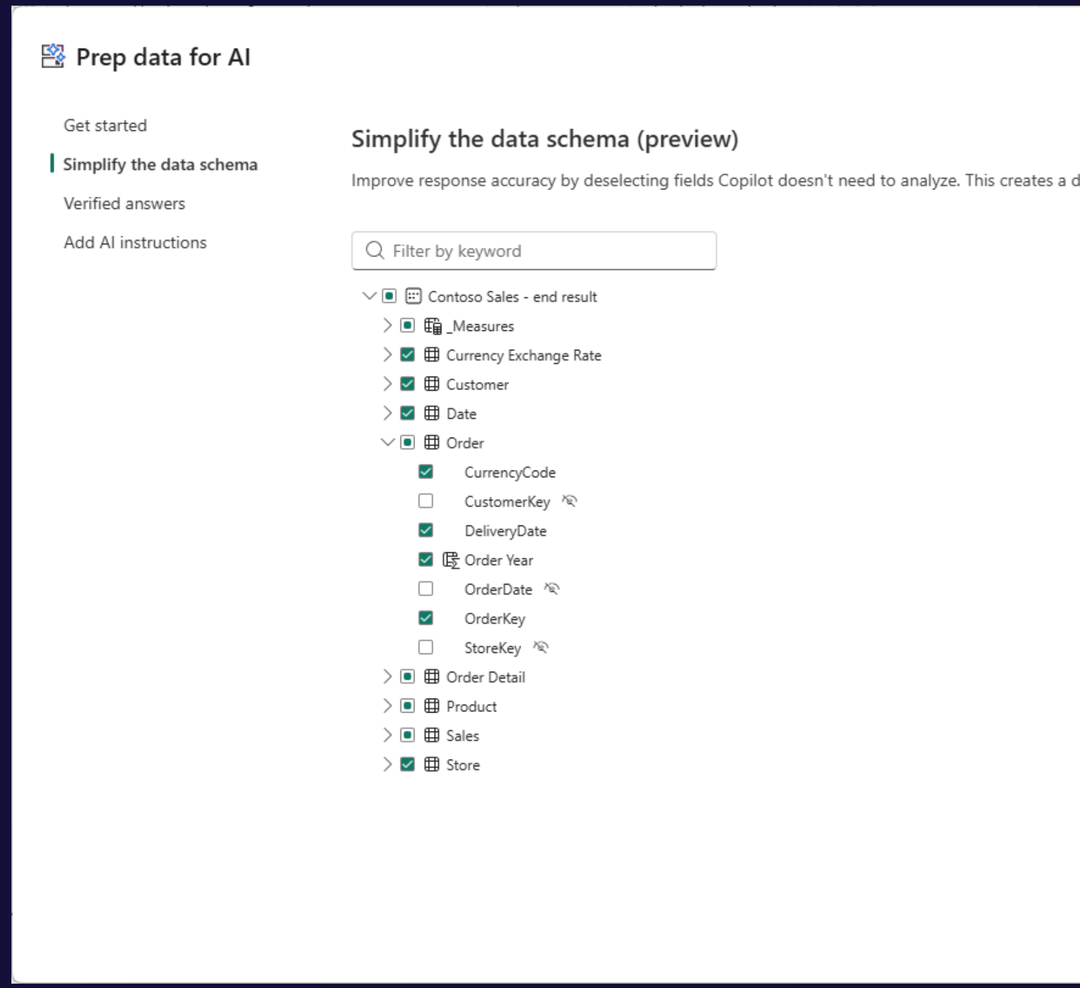


Semantic model only

Semantic models: Prep data for AI – Simplify schema

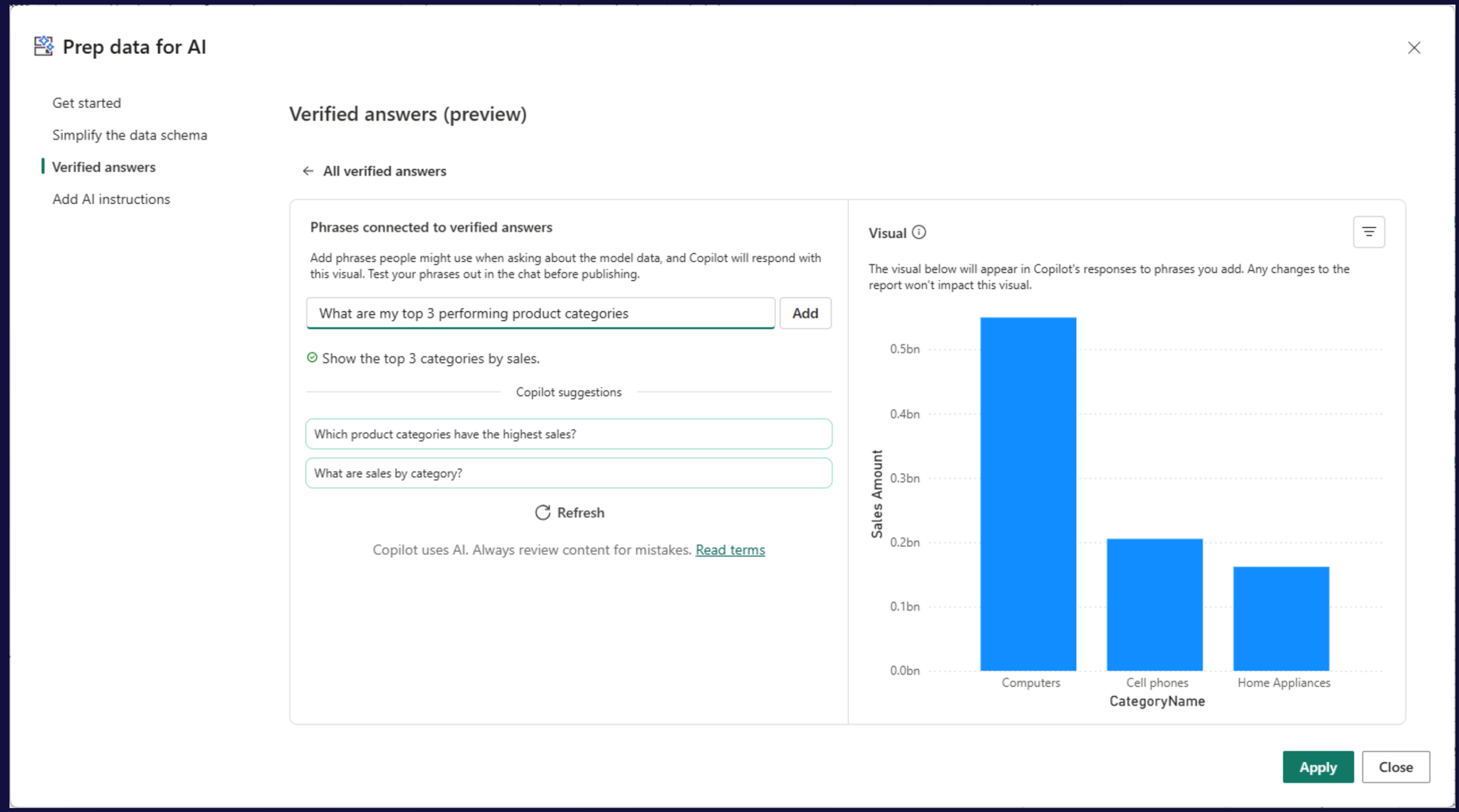
- Only keep what is relevant!
- Hide report-specific measures
  - Conditional formatting
  - Measures outputting texts like visual titles
- Disconnected tables likely cause issues → build relationships!

Keep in mind that Visual Calculations are not part of the semantic model!



Semantic model only

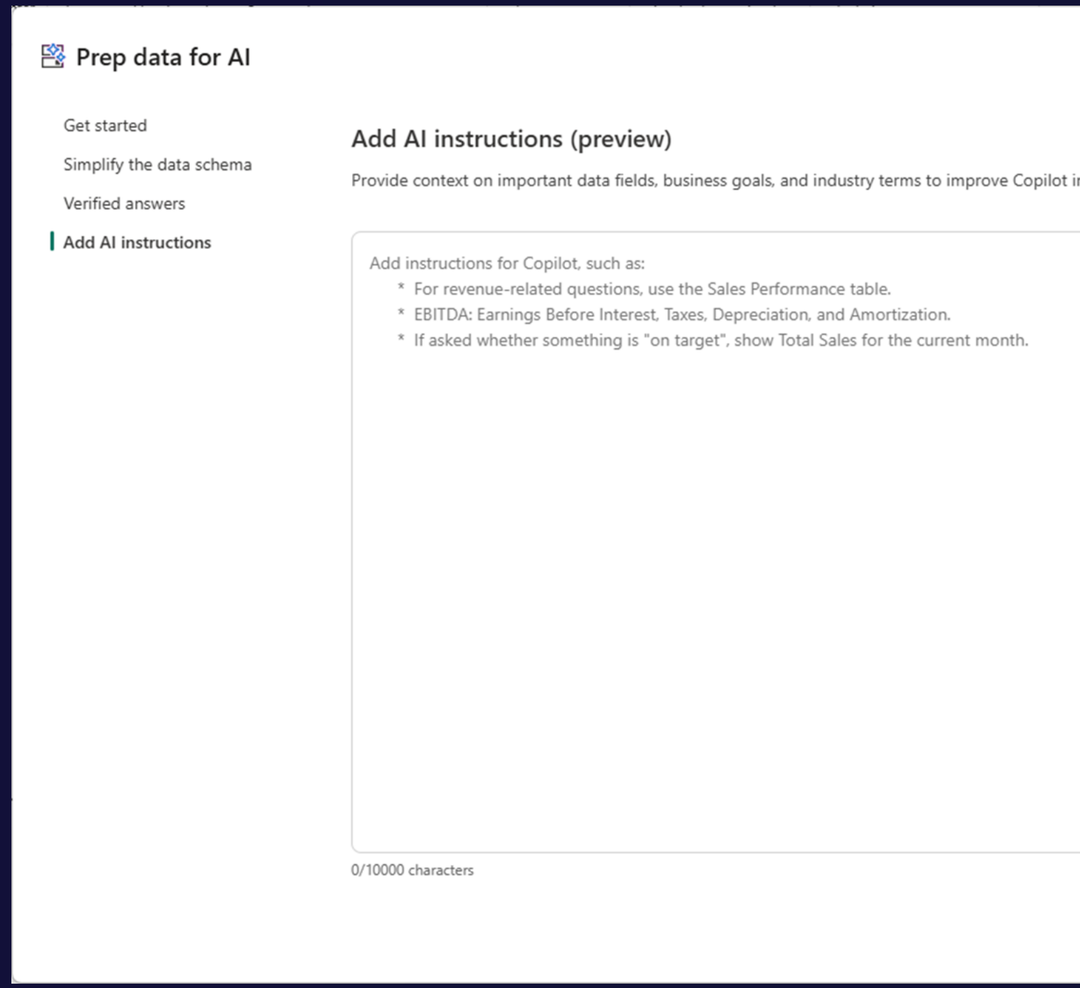
Semantic models: Prep data for AI – Verified answers



Semantic model only

Semantic models: Prep data for AI – AI instructions

- Unstructured guidance (textual)
- No guarantee the guidance is followed exactly – see it as a hint!
- Response formatting
  - Style
  - Tone
- Business specific logic – for example:
  - Define what is a top performer – high or low?
  - Peak / off-season: which months are these?



Common pitfalls

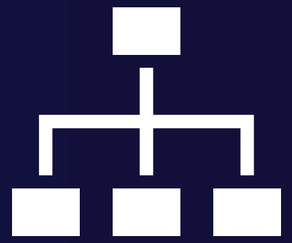
- Conflicting or overly **complex** instructions
- Skipping **schema refinement** – especially with large semantic models containing many tables
- Not using a **star schema** (resulting in inefficient DAX)
- Relying on **hidden fields**
- Including **unnecessary measures**: Simplify the schema,
  - Remove measures that are not relevant or broken,
  - Hide report measure (for conditional formatting for example),
  - Avoid duplicating logic in different measures
- **Non-descriptive naming** (technical names for objects)
- Relying on **implicit measures**: high chance of inconsistency!
- **Ambiguous date fields** (e.g. Order Date, Ship Date, …)



Advantages of using a Semantic Model as a data source



Allows metadata enrichment



Business logic (measures)



User friendly and terminology



Single source of truth (incl reports)

Data Agent SDK

- **Programmatic Management:** Create, update, and delete Data Agent artifacts seamlessly.
- **Data Source Integration:** Easily connect to and integrate multiple data sources, for enhanced data analysis and insight generation.
- **OpenAI Assistants API Support:** Use the OpenAI Assistants API for rapid prototyping and experimentation.
- **Workflow Automation:** Automate routine tasks to reduce manual efforts and improve operational efficiency.
- **Resource Optimization:** Optimize configuration and management of Data Agent resources to better align with specific customer needs.

Copy

```
%pip install fabric-data-agent-sdk
```

Keep in mind!

The answers will only be as good as your data!

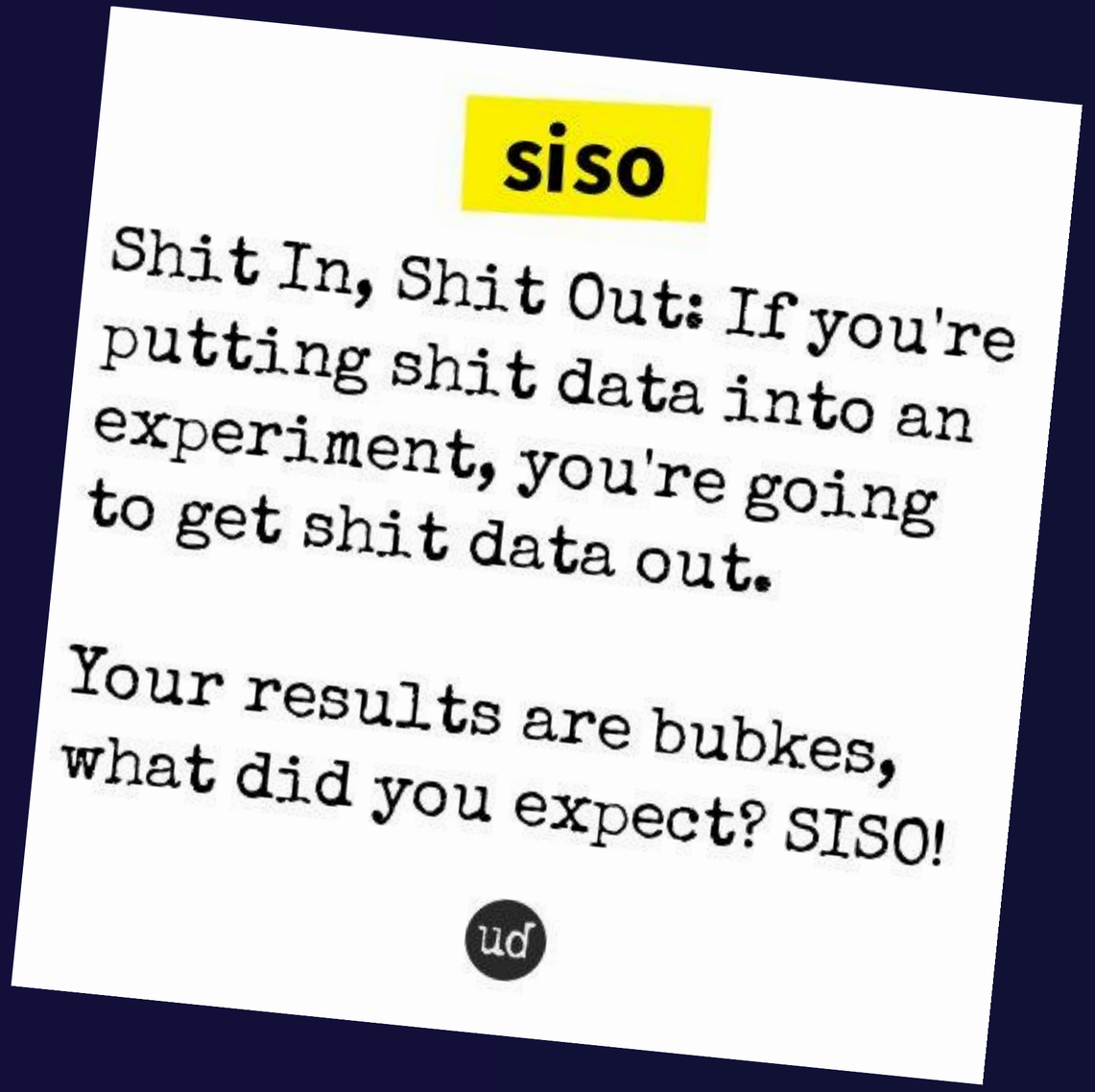


Image from Urban Dictionary: https://x.com/urbandictionary/status/958044898212630528

Diagnostics

“AI is wrong” to structured root cause analysis!

Where to start your investigation when the answer is not what you’re looking for?

Permissions

- RLS / CLS
- Item level access
- governance policies

Prompt & grounding

- Clear question
- Proper metadata

Source grounding

- Semantic model, lakehouse, warehouse, eventhouse
- Relationships and item naming

Notes for model

- Tone & voice
- Context (what is a good performer, high or low?)
- Hints (example queries)

Generated query analysis

- Diagnostics

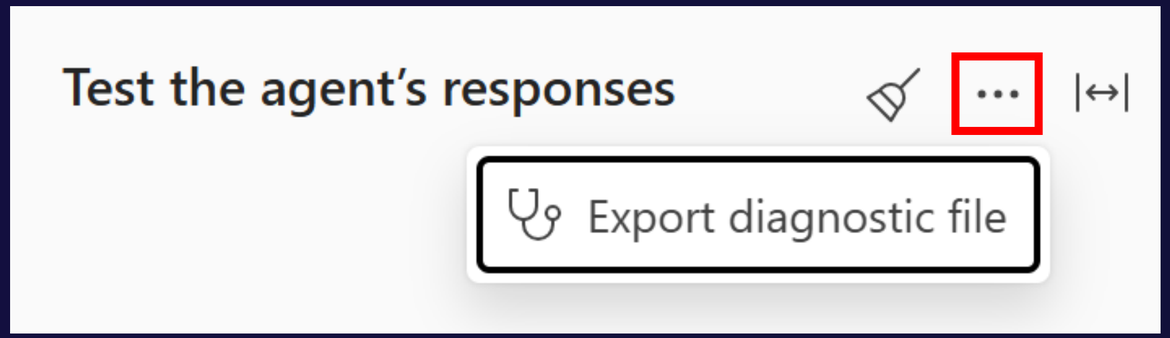
Capacity & performance

- Storage mode (for semantic models)
- Model size
- Load on the capacity

Diagnostics

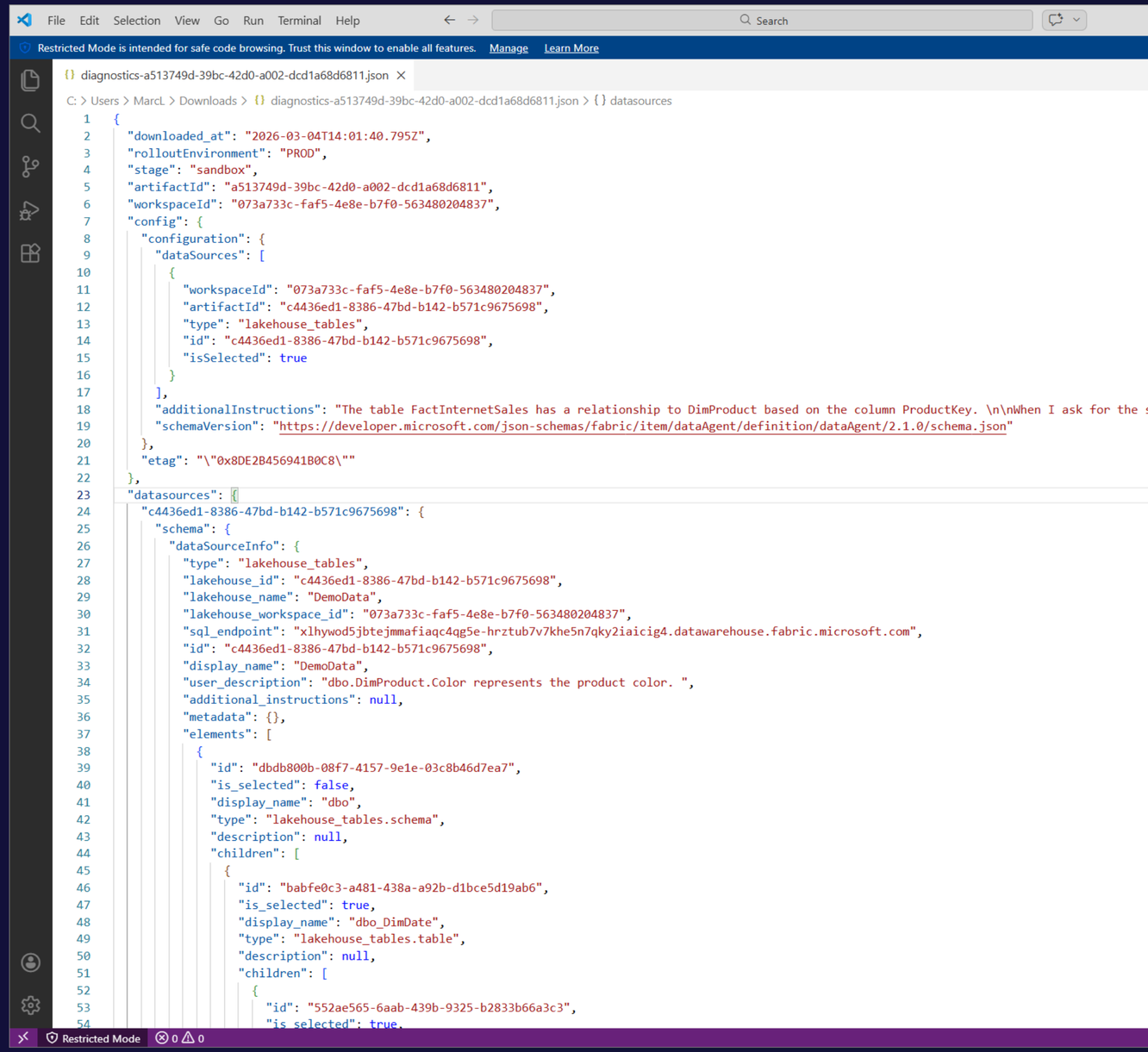
“AI is wrong” to structured root cause analysis!

Where to start your investigation when the answer is not what you’re looking for?



- User prompts
- Data source schemas and configurations
- File metadata and contents
- Environment and artifact identifiers (GUIDS)
- Any proprietary information you may have entered

Demo
Data Agent Diagnostics

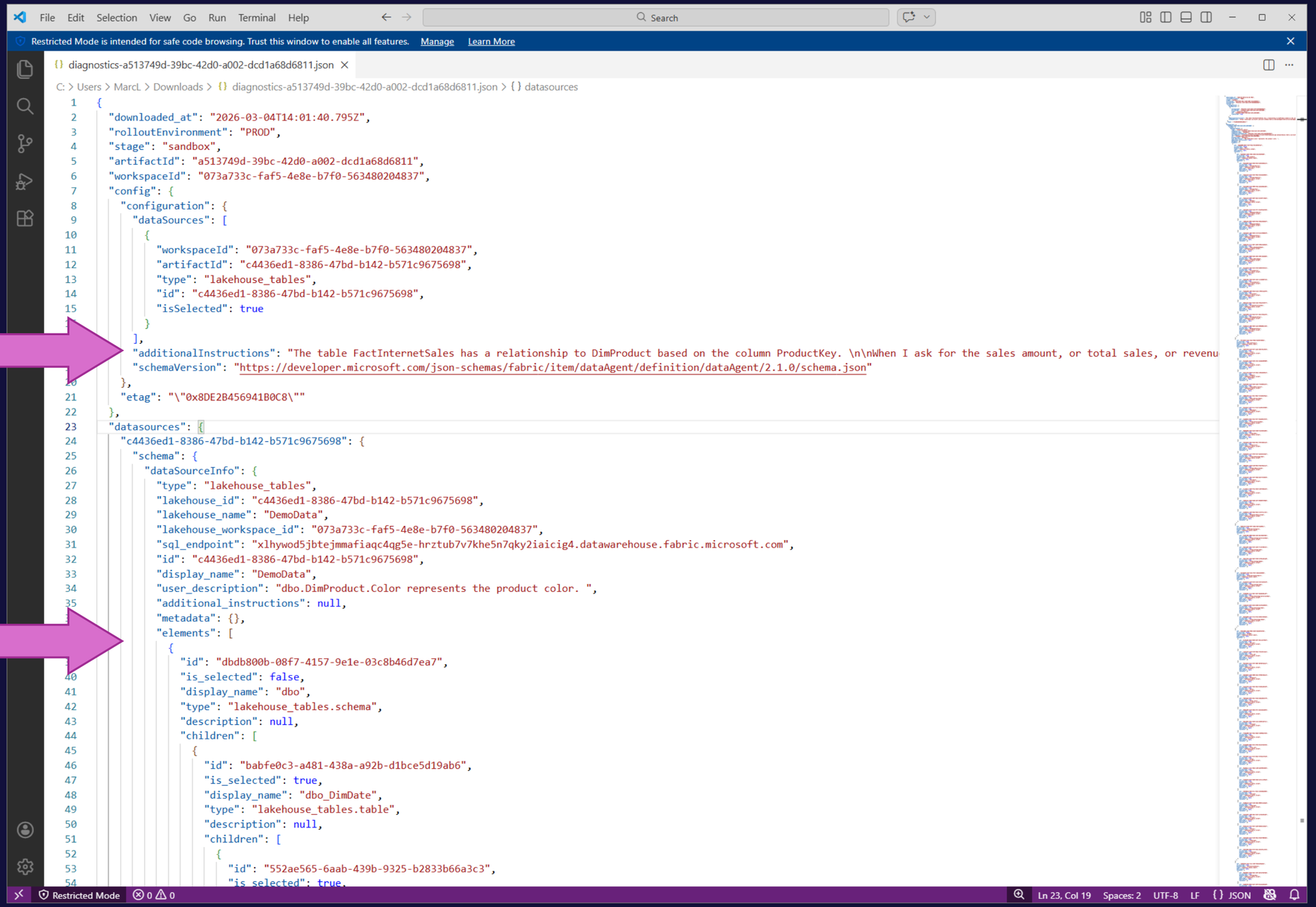


Diagnostics

Source instructions

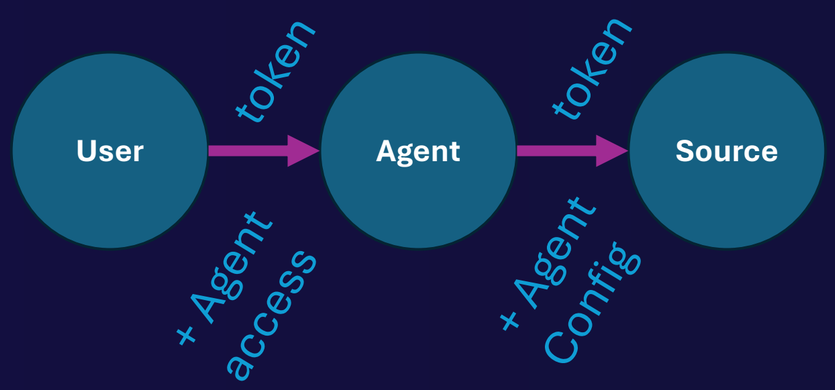
Schema information

...and much more!



Permissions

Data access



Pass-through mechanics

- User credentials flow to the data source
- Permission evaluated per-user, per-query
- RLS/OLS etc. are baked-in

Service principals (SPN) are **NOT** supported, Access requires a user identity.

Data source permission requirements

Semantic model

Read permissions only! *(previously also build)*

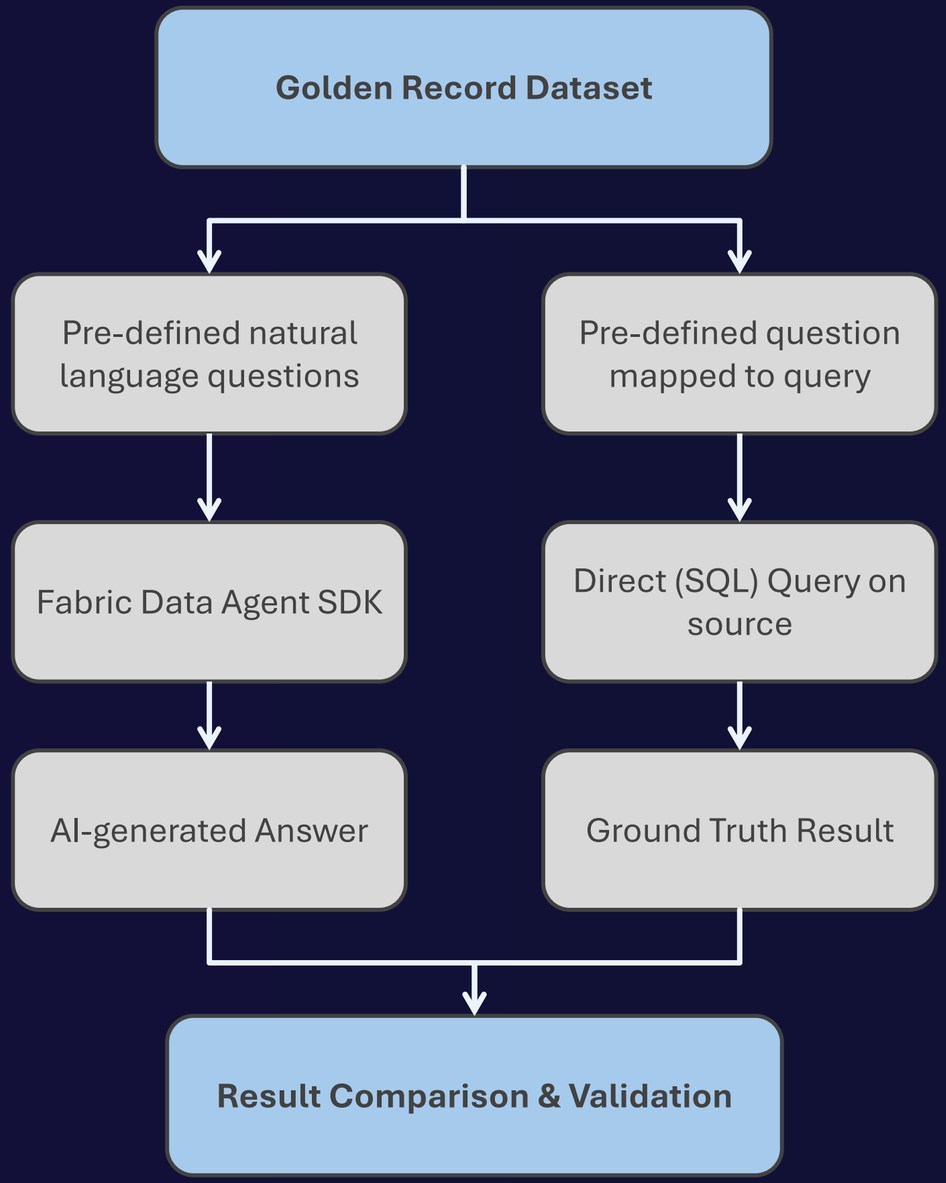
Agent access

Permission	Description	View Config	Edit Agent
Default (None)	N/A	❌	❌
View details	View the configuration and settings, but make no changes.	✅	❌
Edit and view details	Edit the configuration and settings.	✅	✅

Golden record testing

- Establishes trust by validating your Data Agent against a known, authoritative dataset.
- Results are compared one-to-one
- Enables objective evaluation

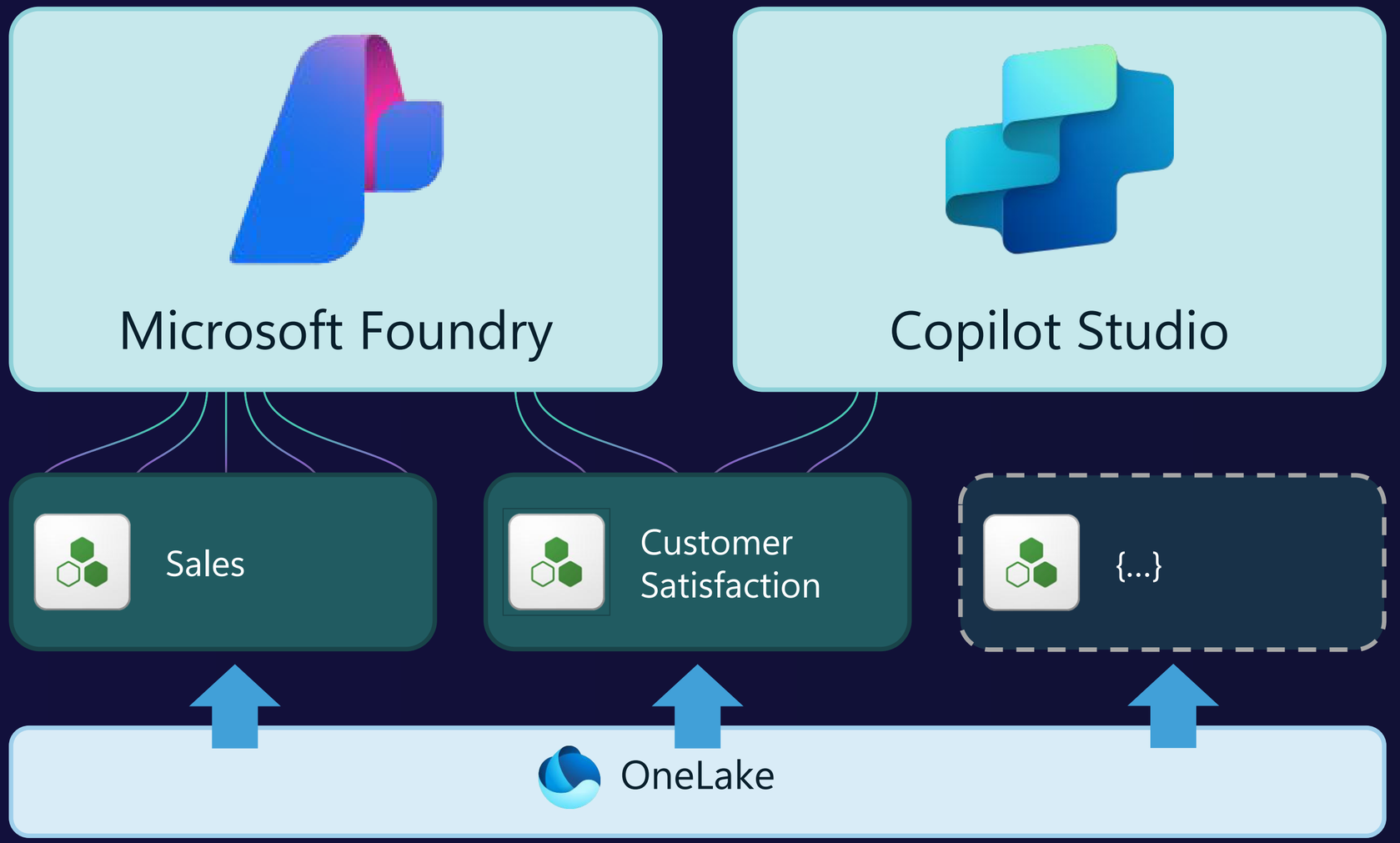
Gains confidence in the data agent, produces accurate, explainable and repeatable answers



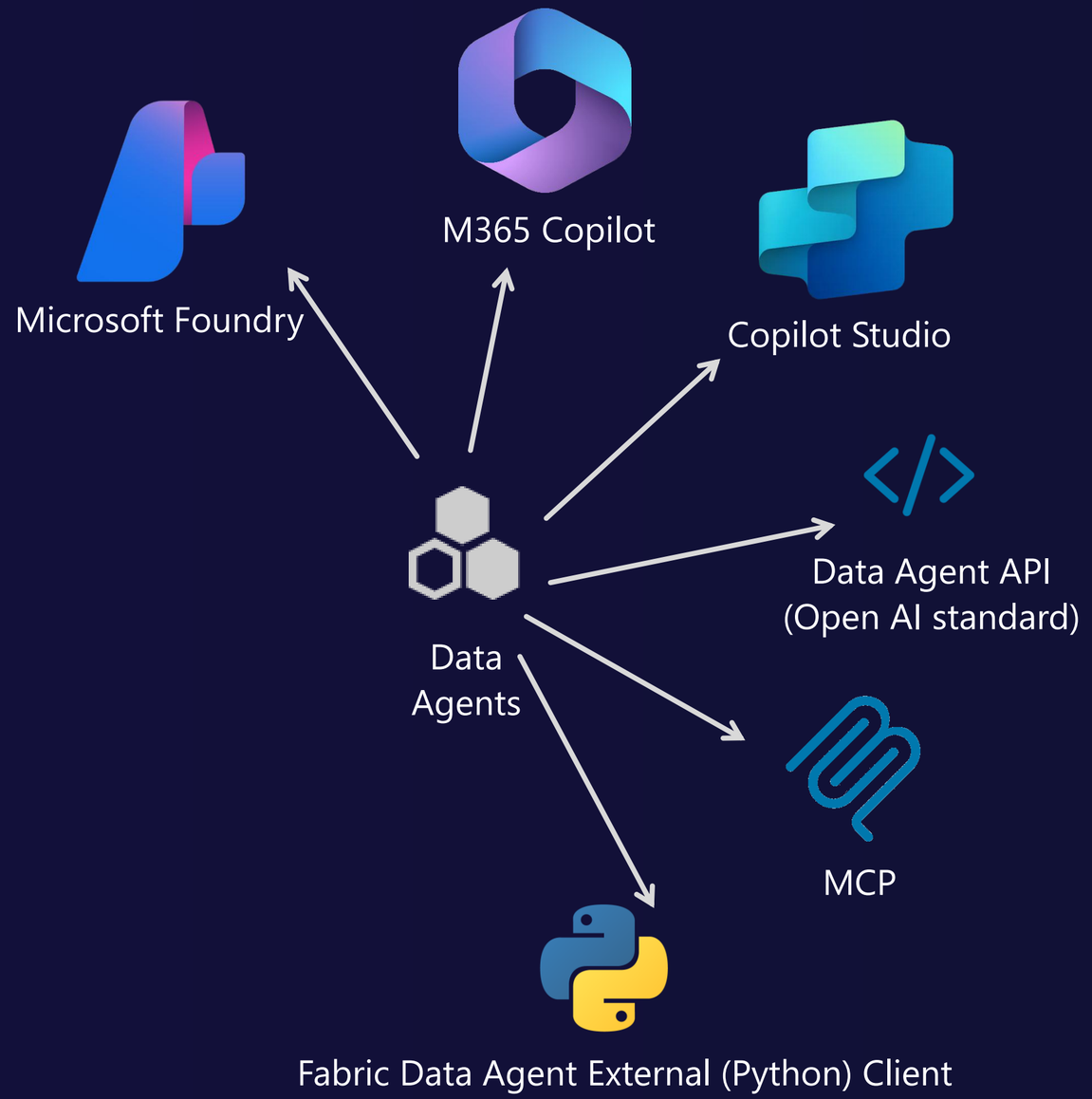
Demo

Agent Architecture

One to rule them all?



Part of a larger **ecosystem**



Recipe to successful implementation



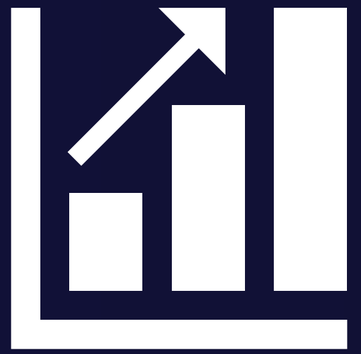
Reliable answers

Through decent preparation & (automated) testing



Secured

Inheritance and passthrough of permissions



Well adopted

Measured and tested with diagnostics

Wrap-up

- Fabric Data Agents enable **natural, governed access** to data across OneLake, semantic models, lakehouses, warehouses, and eventhouses.
- **Strong foundations matter most**, with well-prepared schemas, verified answers, and clear AI instructions driving accuracy and trust.
- **Security and diagnostics are essential**, ensuring correct permissions, explainable behavior, and structured root-cause analysis when results differ.
- Data Agents fit into **a broader ecosystem**, integrating with Fabric, Power BI, Copilot Studio, Microsoft Foundry, and custom applications.

Key takeaway:
Reliable AI answers come from reliable data, thoughtful design, and continuous validation.

Thanks for attending!



Marc Lelijveld
Architect & Technical Advocate
Macaw Netherlands

Microsoft® Most Valuable Professional

linkedin.com/in/MarcLelijveld

Data-Marc.com

Mastering Fabric Data Agents: From Setup to Success

Session Evals

These slides + the code
https://github.com/marclelijveld/Slide-decks
https://github.com/marclelijveld/fabric-automation